MARIAGIULIA MARTINELLI

Bioarchaeology | Palaeogenetics | Conservation Science

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PROFILE

I study what happens to biological material after deposition: the taphonomy of human skeletal remains, the microbial biodeterioration of stone, and the survival of ancient biomolecules. I work across **bioarchaeology**, **palaeogenetics** and **conservation science**, from osteological and laboratory practice through to reproducible, openly released R pipelines.

EDUCATION

Master's Degree in *Science and Technology for the Conservation of Cultural Heritage* (LM-11)
Final grade: 110/110 cum laude

***Sapienza* University of Rome**

09/2024 – 10/2026

- Thesis title: ***Conservation and Reassessment of an Overlooked Skeletal Collection: Historical and Bioarchaeological Study of Human Crania from Etruscan Territory Acquired by G. Sergi in the Early 20th Century***.
- Thesis supervisors: Dr. Ileana Micarelli, Prof. Mary Anne Tafuri, Prof. Laura Maria Michetti
- Programme delivered in Italian and English; admission conditional on assessed B2 English (CEFR).
- Student Representative for the Department of Environmental Biology (Bachelor, Master and PhD level) and for the LM-11 Master's Degree Class.

Erasmus+ Exchange Student

Stockholm University, Sweden

01/2026 – 06/2026

- Courses: *Palaeogenetics* (BL7069); *Modelling Tools for Environmental Scientific Studies* (MI4011); *Communicating Environmental Science* (MI4008).

Bachelor's Degree in *Tecnologie per la Conservazione e il Restauro dei Beni Culturali* (L-43)
Final grade: 110/110

***Sapienza* University of Rome**

09/2021 – 12/2024

- Thesis title: ***Studio dell'evoluzione del microbiota sulle pareti del Casino Algardi (Roma) in seguito ad interventi conservativi con prodotti sostenibili***.
- Thesis supervisors: Prof. Enea Gino Di Domenico, Dr. Chiara Alisi, Prof. Viviana Fonti, Prof. Claudio Chimenti
- Student Representative of the L-43 Degree Class.

RESEARCH EXPERIENCE

Research Internship at the *Centre for Palaeogenetics* (CPG)

Stockholm University / *Sapienza* University of Rome

04/2026 – 06/2026

- Analysed 250 ancient DNA samples from Roman, Visigothic, Islamic and Medieval necropolises in Valencia (Spain), sampled from petrous bone, teeth and phalanges.

- Ran Kraken2 (GTDB) and KrakenUniq (MicrobialNT) classification pipelines as SLURM job arrays on the Dardel HPC cluster (PDC, KTH). Assessed cross-tool agreement (Procrustes, Spearman, PCoA on Bray–Curtis) and screened all samples for *Yersinia pestis*, *Mycobacterium tuberculosis* and *Tannerella forsythia*.

Osteological Laboratory, *Tusculum* Project

EEHAR-CSIC (Escuela Española de Historia y Arqueología en Roma)

12/2025 – 01/2026

- Catalogued and analysed medieval human remains from the Area G cathedral site (DG-ABAP no. 245): skeletal inventory and element identification, biological profile, and macroscopic recording of preservation, weathering and taphonomic alteration.
- Managed the digitisation of the anthropological dataset, ensuring traceability between remains, records and photographic documentation.

Research Internship at the *Palaeoanthropology and Bioarchaeology Laboratory*

***Sapienza* University of Rome**

07/2025 – 12/2025

- Bioanthropological study and critical re-evaluation of a human cranial collection acquired by Giuseppe Sergi (early 20th century): biological profile, dental palaeopathology, history of previous study, and systematic assessment of the state of conservation, distinguishing taphonomic damage from damage introduced during excavation and curation. The objective was to revitalise a neglected scientific heritage collection and propose new avenues for research.
- Designed the reproducible R pipeline used for the collection (*Etruscan-Analysis*).

Research Internship at the ENEA Research Centre, Casaccia (Rome)

ENEA Casaccia

04/2024 – 12/2024

- Designed and executed independently an end-to-end metagenomics workflow on microbially mediated deterioration of a mineral substrate: field sampling and photographic documentation of biofilms on the historic walls of the *Casino del Bel Respiro*, genomic DNA extraction, PCR (16S rRNA, ITS), amplicon sequencing and phylogenetic classification.
- Built the R pipeline from scratch (phyloseq, vegan, ggplot2): alpha- and beta-diversity, taxonomic composition and functional-group succession under sustainable conservation treatments.

Research Internship: Taxonomic Identification and Iconographic Analysis

***Sapienza* University of Rome**

2022 – 2024

- Study of *The Four Elements* cycle by Jan Brueghel the Elder through taxonomic classification of the species represented.

AWARDS AND RECOGNITIONS

Tutoring Fellowship (Cat. B1), Degree Programmes in *Scienze Applicate ai Beni Culturali*

Department of Environmental Biology, *Sapienza* University of Rome

a.y. 2026/2027

- Awarded through public comparative selection (Call BT-B1 18/2026).
- 75 hours of teaching and laboratory support for the L-43 and LM-11 degree classes (supervision: Prof. M. Beccaccioli, Prof. L. Medeghini, Prof. A. Ciccola).

Merit Scholarship – Academic Excellence Track (Master's)

***Sapienza* University of Rome**

03/2026 – 08/2026

- Highly selective merit-based scholarship and honours programme (**Percorso d'Eccellenza**) for top-ranked students, requiring an integrative curriculum of advanced research beyond standard degree requirements.

- Ancient metagenomic analysis of Roman-Medieval archaeological samples, at the *Centre for Palaeogenetics*, Stockholm.

Merit Scholarship – Academic Excellence Track (Bachelor's)

***Sapienza* University of Rome**

04/2023 – 12/2024

- Highly selective merit-based scholarship and honours programme (**Percorso d'Eccellenza**) for top-ranked students, requiring an integrative curriculum of advanced research beyond standard degree requirements.
- Iconographic and taxonomic analysis of the *Four Elements* cycle by Jan Brueghel the Elder.

Student Assistantship, Earth Science Library

***Sapienza* University of Rome**

2023 – 2025

- Merit-based student collaboration grant awarded through competitive University ranking.

PUBLICATIONS

C. Alisi, V. Fonti, M. Martinelli, G. Migliore, P. Paganin, F. Tasso
Abstract in the Proceedings of *The Omics & Heritage (O&H)*, Rome, Italy, 14–15 May 2024
Unveiling the Microbial Hazard: Exploring Biofilm Composition on Casino Algardi for Conservation Insights

MANUSCRIPTS IN PREPARATION

M. Martinelli, V. Fonti, P. Paganin, F. Tasso, C. Alisi
International Biodeterioration & Biodegradation (manuscript in preparation, 2026)
Microbial Biodiversity on the Casino del Bel Respiro Walls: Implications for Conservation Strategies

L. Morucci, A.A. Bastanza, M. Martinelli, S. Iorio
International Journal of Paleopathology (manuscript in preparation, 2026)
A Systematic Review of Analytical Methodologies in Bioarchaeology of Stress in Non-Adult Populations: Recommendations for the Next 50 Years

M. Martinelli, C. Chimenti
Manuscript in preparation, 2026
The Animal World in Jan Brueghel's Four Elements

ORAL CONTRIBUTIONS

Orizzonti Antichi tra Ordine Perduto e Nuove Armonie
Oral contribution *Glycyrrhiza glabra vs Biofilm: nuove prospettive per la conservazione dei beni culturali*
M. Martinelli, V. Fonti, F. Tasso, P. Paganin, C. Alisi; University of Pisa · 02–04/12/2025

The Omics & Heritage (O&H)
Co-author of oral contribution *Unveiling the Microbial Hazard: Exploring Biofilm Composition on Casino Algardi for Conservation Insights*
C. Alisi, V. Fonti, **M. Martinelli**, G. Migliore, P. Paganin, F. Tasso; CNR, ISB · 14–15/05/2024

OPEN SOURCE PROJECTS

Etruscan-Analysis: Bioarchaeological Pipeline (R), ***to be released***

Bioarchaeological-Systematic-Review (R)

kraken2-krakenuniq-comparison (R, Shell)

CO_2 Time Machine (JavaScript)

scienze-abc (HTML)

Biofilm Metagenomics Workflow (Casino Algardi), ***to be released***

SKILLS

Osteology, Taphonomy & Conservation of Skeletal Remains
Skeletal inventory and element identification · biological profile (age, sex, stature) · dental palaeopathology · macroscopic recording of preservation, weathering and taphonomic alteration · selection of elements and sampling loci · collection management, photographic documentation and record–specimen traceability

Laboratory & Analytical Practice
Genomic and ancient DNA extraction · PCR (16S rRNA, ITS) · amplicon sequencing · sterile technique and contamination control in low-copy-number contexts · optical and stereomicroscopy of heritage and biological materials · biofilm sampling

Data Analysis, Statistics & Reporting
R (tidyverse, phyloseq, vegan, ggplot2, sf, leaflet) · non-parametric testing · multivariate ordination (PCoA, Bray–Curtis, Procrustes) · reproducible pipelines, Git/GitHub, LaTeX · publication-quality figures, reports and manuscripts

Bioinformatics & High-Performance Computing
Kraken2 · KrakenUniq · read-count-based pathogen screening · SLURM job arrays on HPC (Dardel, PDC/KTH) · shell scripting · Linux command line

Core Scientific Background
General, organic and analytical chemistry · human anatomy and skeletal biology · microbiology and biology of deterioration · biostatistics

Communication, Organisation & Autonomy
Independent project management from sampling design to written report · elected student representative across three degree levels (L-43, LM-11, PhD) · conference presentations and co-authored manuscripts

LANGUAGES

Italian
Native speaker (C2). Full academic and professional use.

English
B2 (CEFR), assessed at admission to the LM-11 Master's programme, which is delivered in Italian and English. Full Erasmus+ semester at Stockholm University.

CONFERENCES, WORKSHOPS & WEBINARS

International Conference on Palaeogenomics (ICP2026)
Stockholm, Sweden · 23–26/06/2026

Webinar: ***Bottleneck o Consanguineità: come si entra in Accademia***
GIGA (Gruppo Italiano Giovani Antropolog3) · 15/12/2025

Webinar: ***Starter Pack per la ricerca: dalla scrittura della tesi ai primi congressi***
GIGA (Gruppo Italiano Giovani Antropolog3) · 26/11/2025

Seminar: ***Nuove opportunità professionali per i biologi nel mondo dell'arte e dei beni culturali***
FNOB, FIB · 27/02/2025

Webinar: ***Primi passi con R: introduzione al linguaggio per analisi statistiche***
Consortium GARR · 12/06/2024

SAGacia24: ***Primo Simposio Arte che Guarda alla Scienza***
Sapienza University of Rome · 05/04/2024

Workshop: ***Dagli Acetali, ai DES, ai FAME: i nuovi solventi green introdotti da YOCOCU nel settore dei Beni Culturali***
YOCOCU · 29/01/2024

Seminar: ***La meteorologia moderna e il ruolo guida dell'Italia nel suo sviluppo storico***
CREA, Università della Tuscia · 29/11/2023

WORK EXPERIENCE

Library Assistant

Library of the Department of Earth Sciences, *Sapienza* University of Rome

01/2023 – 01/2025

Private Science Tutor

Self-employed

2021 – 2023

Tennis Instructor

***Reschio Hotel* Summer Camp**

06/2021 – 08/2021

CERTIFICATIONS

Open Badge: Student Representation
Issued by Bestr · 2025

Open Badge: Service Learning
Issued by Bestr · 2024

Course: *D.Lgs. 81/2008 Health and Safety Training*
KPMG Advisory · 27/02/2023

Certificate of Excellence: Localizing the Sustainable Development Goals (MUSTER)
Issued by Dr. Karl-Ludwig Schibel, President of the Fiera delle Utopie Concrete, Liceo Scientifico "Plinio il Giovane" · 2018

REFERENCES

Dr. Ileana Micarelli
@ *Sapienza* University of Rome

Dr. Chiara Alisi
@ ENEA Casaccia

Dr. Gonzalo Oteo
@ Centre for Palaeogenetics, Stockholm University

Prof. Enea Gino Di Domenico
@ Istituti Fisioterapici Ospitalieri

Autorizzo il trattamento dei dati personali contenuti nel mio curriculum vitae ai sensi del Regolamento UE 2016/679 (GDPR) e della normativa italiana vigente. Acconsento inoltre a che la presente candidatura sia presa in considerazione nei 12 mesi successivi alla selezione anche per ulteriori posizioni con profilo professionale simile.

Signature: ***Mariagiulia Martinelli***

Date: October 22, 2026